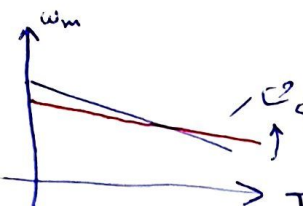


1)  ω_m vs T

الف) $\downarrow V_t \leftarrow \downarrow E_a \leftarrow \downarrow \phi \leftarrow AR$ (1)
 ب) $\uparrow \phi \leftarrow \uparrow I_f \leftarrow \uparrow V_f$ (1)
 ج) $a = P = 4, Z = 300 \rightarrow E_a = \frac{ZP}{2\pi a} \phi \omega = 72/90 \text{ V}$ (1.5)

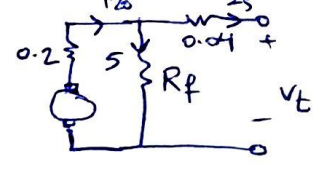
ت) زیرا برای تولید ولتاژ به ϕ نیاز است و در ابتدا ماشین (ترینال) ولتاژ ندارد تا جریان تحریک را فراهم کند پس این جریان اولیه باید توسط شار اولیه (سخت‌اند) مستقل از ولتاژ تأمین شود (1)

2) $\frac{dL_{ss}}{d\theta} = \frac{dL_{rr}}{d\theta} = \frac{-60}{1000} \sin 2\theta, \frac{dL_{sr}}{d\theta} = \frac{-240}{1000} \sin 4\theta \rightarrow T = b^2 (-60 \sin 2\theta - 240 \sin 4\theta) \times 10^{-3}$ (0.5)
 $\rightarrow \theta = 15^\circ \rightarrow T = -15.22 / -8.56 \text{ N.m}$ (1)

ب) $\theta = \omega t \rightarrow \bar{T} = \frac{-b^2}{1000} (60 \sin 2\omega t + 240 \sin 4\omega t) = 0$ چون ضرب t در هیچ یک از جملات برابر صفر نیست (1)
 Total: 5

3) الف) $P_{out} = T_{out} \times \omega = 75 \times 1200 \times \frac{2\pi}{60} = 9424.78 \text{ W}$ $P_{in} = P_{out} + P_{loss,elec} + P_{rot}$ (1)
 $P_{loss,elec} = 1.1 \times C^2 = 990 / 1126.4 \text{ W} \rightarrow P_{in} = 10964.78 / 11101.18 \text{ W}$ (0.5)
 $\rightarrow \eta = \frac{P_{out}}{P_{in}} = 85.96\% / 84.9\%$ ب) $V_t = \frac{P_{in}}{c} = 365.49 \text{ V} / 346.91 \text{ V}$ (1)
 Total: 4

4) $\frac{T_2}{T_1} = \frac{\omega_2}{\omega_1}, \frac{T_2}{T_1} = \left(\frac{I_{a2}}{I_{a1}}\right)^2 \rightarrow \frac{d}{800} = \left(\frac{I_{a2}}{20}\right)^2 \rightarrow I_{a2} = 18.03 \text{ A} / 16.58 \text{ A}$ (1)
 $E_{a1} = 230 - 1 \times 20 = 210 \text{ V}, \frac{E_{a2}}{E_{a1}} = \frac{\phi_2 \omega_2}{\phi_1 \omega_1} = \frac{I_{a2}}{I_{a1}} \times \frac{\omega_2}{\omega_1} = \frac{18.03}{20} \times \frac{650}{800} \rightarrow$ (0.5)
 $E_{a2} = 153.82 \text{ V} / 119.69 \text{ V} \rightarrow 1 + R_c = \frac{230 - E_{a2}}{I_{a2}} \rightarrow R_c = 3.23 \Omega / 5.65 \Omega$ (1)
 Total: 3.5

5)  $I_f(\text{eff}) = 5 + \frac{20}{1000} \times I_s - e, I_s = 120 - 5 = 115 \text{ A}$ (0.5)
 $\rightarrow I_f(\text{eff}) = 6.6 \text{ A} / 6.5 \text{ A}$ (0.5) $\frac{E_a - 216}{6.6 - 4.8} = \frac{240 - 216}{6.8 - 4.8}$ (0.5)
 $\rightarrow E_{a0} = 237.6 \text{ V} / 236.4 \text{ V} \rightarrow E_a @ 1150 = \frac{1150}{1000} \times E_{a0} = 273.24 \text{ V} / 271.86 \text{ V}$ (1)
 $\rightarrow V_t = E_a - 0.2 \times 120 - 0.04 \times 115 = 244.64 \text{ V} / 243.26 \text{ V}$ (1)
 $\rightarrow R_f = \frac{E_a - 0.2 \times 120}{5} = 49.85 \Omega / 49.57 \Omega$ (1)
 Total: 4.5